Fine-Grained Binary Object Segmentation in Remote Sensing Imagery via Path-Selective Test-Time Adaptation

Kyungsu Lee, Haeyun Lee, Juhum Park, *Member, IEEE*, and Jae Youn Hwang, *Member, IEEE*

***Abstract*— For several decades, the significance of geospatial object segmentation in remote sensing (RS) images has been emphasized for both scientific and industrial purposes. Object segmentation plays a pivotal role in the analysis of urban and rural area expansion, as well as in advancing sustainable development within the realm of RS. Deep learning (DL)-based segmentation methodologies, overcoming the limitations of the conventional vision-based analysis, have yielded precise predictions by utilizing convolutional neural networks (CNNs). However, CNNs classify images at the pixel level and generate outputs based on probability distributions derived from the SoftMax function. This approach precludes the reflection of morphological properties, such as shape and object density, during predictions in RS imagery, leading to imprecise results. In addition, due to the intrinsic attributes of probability-based segmentation, fine-grained segmentation may not be achieved, leading to coarse predictions in the boundaries of geospatial objects. To address this issue, this article introduces a novel DL framework, the density-based guide network (DG-Net), which incorporates the density of segmentation targets into pixel-wise classification through a test-time adaptation learning methodology. DG-Net first discerns the density of segmentation targets in the input images, then fine-tunes the baseline network to reflect this density, thereby generating precise segmentation outputs. The effectiveness of DG-Net is demonstrated through various multitarget segmentation benchmarks in RS imagery. Experimental results demonstrate the superior performance of the DG-Net in object segmentation when compared to state-of-the-art (SotA) models across numerous aerial image and satellite image datasets.**

***Index Terms*— Fine-grained segmentation, object segmentation, remote sensing (RS) imagery, test-time adaptation.**

I. Introduction

ACCURATE recognition of individual objects holds vital importance in practical applications within the field of remote sensing (RS) such as scene recognition and autonomous driving [1], [2], [3], [4], [5]. For instance, precise object recognition in RS imagery is essential to create high-quality digital maps or detect object changes for urban planning and military applications [6], [7], [8], [9]. Particularly, various deep learning (DL)-based segmentation algorithms have been developed for generating high-quality digital maps [8], [9], [10], [11], [12], [13].

In the early era, DL-based segmentation algorithms have been developed based on fully convolutional networks [14]. Then, the DL models with simple yet effective networks have been introduced [15], [16], [17], [18]. However, the previous DL models have exhibited poor segmentation performance due to simple architecture prompting coarse feature extraction from the complex RS imagery. Therefore, advanced DL models, which can utilize the diverse features of various objects, have been required to provide precise segmentation outputs. Hence, more recently, advanced DL models have been developed to improve segmentation performance for specific applications [10], [12], [19], [20], [21], [22], [23], [24], [25], [26]. Nevertheless, object segmentation in aerial imagery is still challenging due to the indistinct shapes of buildings and the extreme variations in color, rotation, and shape of target objects [7], [27]. A critical aspect often underestimated is density, which quantifies the quantity of units within a defined land area. Density holds significant importance in the context of object segmentation within aerial imagery. Nevertheless, contemporary DL models frequently disregard density in their feature extraction process, as they rely on pixel-wise classification and probability-based SoftMax outputs to generate predictions. To alleviate the limitations, density-based segmentation algorithms have been studied [28], [29], [30]. However, although the previous density-based segmentation algorithms have indeed shown improved performance for object segmentation, they have still exhibited limited performance for object segmentation since they used machine learning (ML)- or clustering-based methods without convolutional neural networks (CNNs).

Therefore, we here developed the adaptive DL network, called a density-based guide network (DG-Net) for better object segmentation. DG-Net is designed to incorporate density-related information into feature extraction, facilitating density-oriented segmentation using CNNs. Conceptually related to the concept of test-time adaptation, DG-Net fine-tunes the baseline network with pseudo-labels linked to the

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density of target objects during the segmentation process. The DG-Net consists of two components: a guider and a baseline-network. The guider identifies the density of target objects in input images and subsequently fine-tunes the parameters of the baseline-network via a test-time adaptation learning manner. The fine-tuned baseline-network then predicts precise segmentation maps reflecting the density of targets. For evaluations, the performances of DG-Net were compared to those of other state-of-the-art (SotA) models with various aerial imagery benchmarks. The results demonstrated that DG-Net outperforms other SotA models in binary segmentation. The distinct contributions of our study are highlighted as follows.

1) We formulated the definition of the probability- and density-based segmentation and proposed the DL model, denoted as DG-Net, to improve the defect caused by the probability-based segmentation.
2) For better density-based segmentation, we devised the density-recognizable guider, including novel modules of the attention-based similarity amplifying (ASA)-block, and a similarity assessment (SA)-layer.
3) DG-Net significantly offers much higher intersection over union (IoU) than other SotA models in binary segmentation using multiple benchmarks of RS imagery datasets.

II. Related Works

In the early era of the development of DL, several studies introduced innovative DL-based segmentation methodologies for delineating target objects within aerial imagery. Among these contributions, BSNet demonstrated its capability in extracting multiscale features through the strategic incorporation of multiple pyramid pooling layers [6] into the well-established U-Net framework [16] for precise aerial object segmentation [8]. Gupta et al. [10] on the other hand focused on the segmentation of buildings and roads, leveraging a module capable of exploiting identical information-impacted regions. Wu et al. [11] developed multiconstraint fully convolutional networks and segmented buildings in aerial images. Furthermore, a concerted effort has been made to innovate in attention mechanisms and loss functions to further enhance segmentation performance. Notably, an attention-based model, augmented with a region shuffle attention module and novel layers, was introduced, leading to notable improvements in the segmentation of multiple objects [9], [12]. In addition, the employment of the focal loss [31] emerged as an effective strategy to extract salient features from the intricate and ambiguous landscapes of aerial images [13].

In addition to the conventional neural networks, more advanced DL models have been developed to enhance segmentation accuracy for specific applications. Notably, path-selective networks have been developed to enable approximate paths for input images. Lin et al. [23] introduced the concept of a multipath DL model that effectively utilizes information along the downsampling process of residual connections to achieve better segmentation. Byra et al. [26] on the other hand proposed a U-Net-based model with multichannel and multipath convolutions designed specifically for object segmentation. Zhuang [24] developed a DL model incorporating stacked U-Nets, extracting information through adjacent multipath decoders to enhance segmentation performance. Furthermore, Bai and Zhou [25] leveraged resource constraints and cost-effective upsampling data paths to further improve segmentation accuracy.

More recently, density-based clustering deep-learning models have emerged to enhance segmentation performance for various applications. Han and colleagues introduced an ML algorithm that integrates multiple features, employing mathematical modeling to formulate a density function. This density function was selected for its algorithmic simplicity and was successfully applied to a classification task [29]. On the other hand, Zhang et al. [30] proposed a density-based unsupervised segmentation method, adopting density-based clustering and sensitive parameter setting techniques.

III. Method

A. Density-Based Segmentation for Better Fine-Grained Segmentation

In this study, our principal objective is to precisely delineate target objects (e.g., buildings and roads) in the RS imagery via fine-grained segmentation. Therefore, we conducted a preliminary study of probability-based segmentation methodology and discerned its inherent limitations, which hinder the achievement of the desired granularity in RS imagery segmentation. In the preliminary study, we found that the sequential learning process in probability-based segmentation hinders performance in achieving fine-grained segmentation. These attributes are carried from the diversity of the density of RS imagery. Particularly, different density groups contain different ratios of inner and boundary pixels, leading to the prioritized training, which hampers fine-grained segmentation, for the samples with high density, classified as easy samples. The associated proofs are demonstrated in a supplementary material in detail (see Figs. 14 and 15). To alleviate this limitation, we applied the ensemble learning-based approach for a different density group since the different density ground can effectively alleviate the impact of the ratio in inner and boundary pixels. Suppose the density of the image (I) is subjected to a certain density group (D_i), then the prediction by the DL model (M) is formulated as below

$$M(I) = M_i(I) \quad \text{where } d_I \in D_i \tag{1}$$

where M_i represents the best-performing model for the corresponding density group (D_i). However, in the prediction step, the density cannot be measured since the density value is subject to fine-grained segmentation for target objects in RS imagery. Particularly, the density can be calculated in the fine-grained segmentation map (e.g., ground truth or prediction with 100% accuracy). To this end, a subnetwork (M_{clf}) is designed to initially predict the density of the input image and select the proper subnetwork for the segmentation. The overall network design, which is denoted as

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DG-Net, is formulated as below

$$M_{\text{clf}}(I) = d \implies M(I) = M_i(I). \tag{2}$$

B. DG-Net

DG-Net consists of a guider and a baseline-network. The guider includes classification-network (C-Net) and DG-Siamese-Net, whereas the baseline-network has N baseline-networks, as shown in Fig. 1. The guider predicts a density (see Appendix, Definition 3) index of a target object in an input image. The pipeline of DG-Net is the following: 1) based on the predicted density index, the guider selects a baseline-network optimal for the input image among backbone architectures. 2) The selected baseline-network generates a prediction map, called a semi-prediction (P') in Fig. 1. 3) Then, based on the semi-prediction, the guider is fine-tuned by itself and adaptively fine-tunes DG-Siamese-Net. Here, the gap between the density index (*S.I.*) and the calculated density by using P' is utilized to fine-tune both the DG-Siamese-Net in the guider and the selected baseline-network in the baseline-network, simultaneously. 4) Finally, the fine-tuned baseline-network generates precise predictions.

C. Guider

Since the guider is a similarity-based density predictor, SiameseNetwork [32] is adopted as the baseline architecture of the guider. The detailed layers of the modified SiameseNetwork, denoted as DG-Siamese-Net, are illustrated in Fig. 2(a). To carry out the density index, similarities between an input (I) image and n reference images (R_n) are first calculated. If I is similar to a certain reference image (R_i), I can be then regarded to have the same density index $[d_{\text{idx}}(I)]$ with that of the corresponding reference image $[d_{\text{idx}}(R_i)]$. Then, $d_{\text{idx}}(I) = d_{\text{idx}}(R_i)$. Since the density is not recognizable in the raw input images, I and R_n are individually converted to a class activation map [33] to highlight the target objects in the input images. Here, the pretrained CNNs (C-Net in Fig. 1) are utilized to generate CAMs. By evaluating the similarities between the converted CAMs of I and R_n, the DG-Siame-Net predicts a similarity index between I and each R_i, and then the largest similar $d_{\text{idx}}(R_i)$ is used as the $d_{\text{idx}}(I)$ of I. The detailed description to construct the reference images is illustrated in Section IV-A.

Furthermore, to provide a more accurate density index, the guider includes novel similarity-based modules consisting of an ASA-block and a SA-layer. The schematic architecture of the ASA-block is illustrated in Fig. 2(b). In addition, the schematic architecture of the SA-layer, which is embedded in the ASA-block, is illustrated in Fig. 2(c). The SA-layer measures the similarity between two inputs (feature maps or images). The output of the SA-layer has the same size as that of the input images. Here, each output element is normalized as the range of [0, 1]. The elements indicate pixel-wise similarity scores. A score closer to 1 implies that the inputs have similar spatial information. SA-layer is formulated as follows:

$$\begin{aligned}\text{SA}(I_1, I_2) = \sum_{i=1}^{3} \big[&(Z \circ M)\big(\widetilde{B}(I_1, 2i-1)\big) \\ &\cdot (Z \circ M)\big(\widetilde{B}(I_2, 2i-1)\big)\big]\end{aligned} \tag{3}$$

where $\widetilde{B}(F, i)$ indicates the block local mean of which a filter size is i, and Z and F indicate Z-normalization and min–max normalization operations, respectively. The ASA-block amplifies similarity extracted using SA-layers. Here, two similarity features are generated from an SA-layer while comparing I_1 and I_2, and F_1 and F_2, respectively. The ASA-block amplifies similarity-related features in the middle of convolutional layers by leveraging two features. The ASA-block is calculated as follows:

$$\begin{aligned} S_{I_1,I_2} &= \text{SA}(wI_1 + b, wI_2 + b) \\ \text{ASA}(I_1, I_2, F_1, F_2) &= M\Big(H(S_{I_1,I_2} - \overline{S_{I_1,I_2}}) \cdot \text{SA}(F_1, F_2)\Big)\end{aligned} \tag{4}$$

where H indicates the Heaviside step function and w and b indicate the weight and bias of a convolution filter in an ASA-block, respectively. In addition, $\overline{S}$ indicates the average value of the matrix of all elements of S. Here, since the ASA-block requires convolution output immediately, the ASA-block is placed after the convolution operation in DG-Siamese-Net. Likewise, the convolution operations are replaced by the ASA-blocks in baseline-networks as illustrated in Fig. 2(d). Note that the baseline-network of the baseline-network and DG-Siamese-Net have the same structures as well as share the same parameters.

Fig. 3 indicates the pipeline of the proposed block local mean operation. The block local mean measures the similarity between two input images. The input images are averaged using different sizes of filters while preserving the size of the input images. Then, the cosine similarity is evaluated using the averaged features in the same levels. By adding similarity maps, the similarity between two input images is generated. Here, to rectify the range of the added similarity, normalization techniques are applied.

D. Optimization of DG-Net

In DG-Net, trainable CNNs are 1) C-Nets in a guider, 2) DG-Siamese-Net in the guider, and 3) baseline-networks. Here, C-Nets are pretrained in the pretraining phase using ImageNet and individual benchmark datasets (see Section IV-A). The C-Net is a binary classifier that is optimized for a classification task that determines whether an input image includes target objects or not. The following loss function is utilized for training the C-Net:

$$\mathcal{L}_{\text{cnet}} = \sum_i y^{(i)} \log(\hat{y}^{(i)}), \quad \hat{y}^{(i)} = \text{CNet}(I^{(i)}) \tag{5}$$

where $I^{(i)}$ is the ith sample, *CNet* is the function standing for C-Net, and $y^{(i)}$ is the annotation corresponding to $I^{(i)}$. Here, $y^{(i)}$ indicates 1 if target objects exist in $I^{(i)}$, otherwise, 0, and $\hat{y}^{(i)}$ is the predicted probability in the range of [0, 1] by C-Net. In contrast, baseline-networks and DG-Siamese-Net

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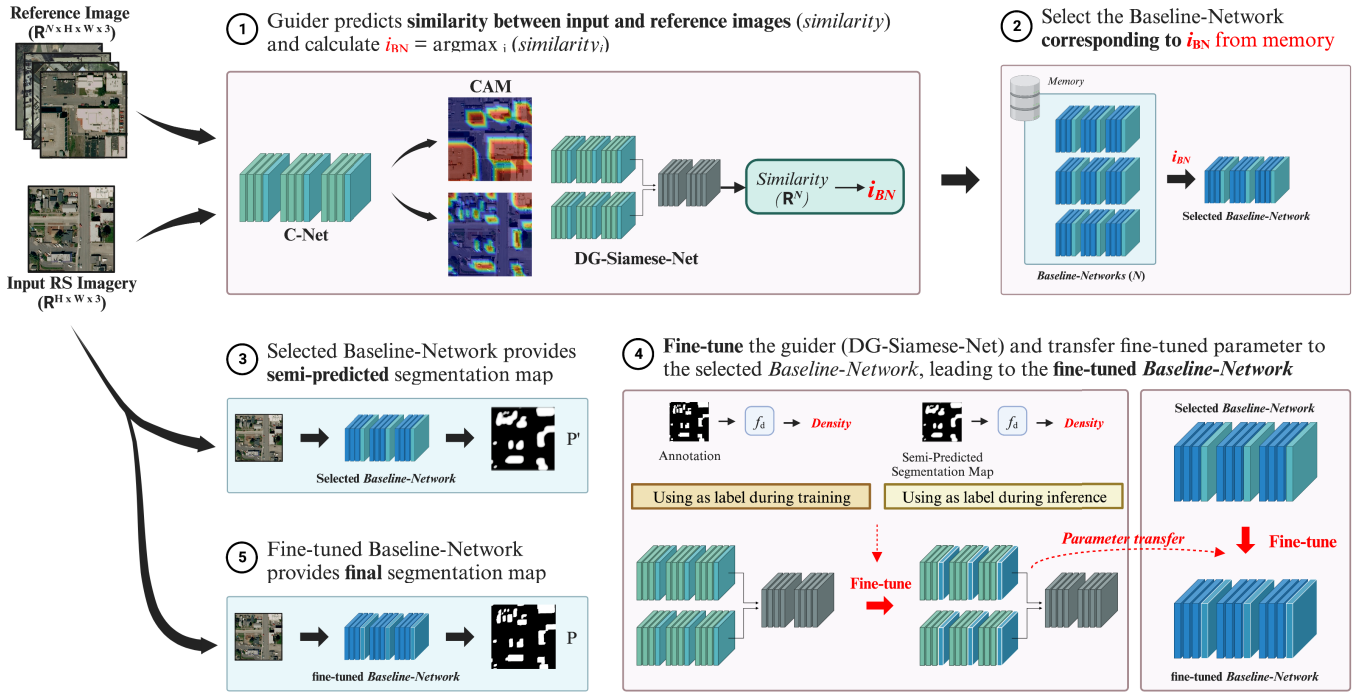

Fig. 1. Graphical abstract of DG-Net. 1) The guider recognizes the density range of target objects in an input image (I). 2) The guider selects the optimal network among baseline-networks based on the density of the I. 3) The selected baseline-network generates a semi-prediction map (P') using I. 4) The DG-Siamese-Net is fine-tuned by calculated density (P') and predicted density (*S.I.*), and the guider fine-tunes the baseline-network based on the density of I. 5) The fine-tuned baseline-network, which is approximate to the density of objects in an input image, generates the final prediction (P).

include convolution filters in the ASA-block [see Fig. 2(b)] are trained in the main training phase. The following equations are utilized to optimize DG-Net:

$$\mathcal{L}_{\text{siam}}(I^{(i)}) = \sum_{i} \sum_{n=1}^{N} \hat{s}\left(I^{(i)}, R_n\right) \cdot \log P_{\text{siam}}(I^{(i)}, R_n) \tag{6}$$

$$\mathcal{L}_{\text{seg}}(I^{(i)}, G^{(i)}, \theta_n) = s\left(I^{(i)}, R_n\right) \sum_{i} \log P^{n}_{\text{seg}}(I^{(i)}) G^{(i)} \tag{7}$$

where $I^{(i)}$, $G^{(i)}$, and R_n indicate the input, corresponding ground truth, and nth reference images. In addition, we assume that $\hat{s}(x_1, x_2)$ is a length-N vector that contains the similarity annotations, and $s(x_1, x_2) = 1$ whenever x_1 and x_2 are in the same class (density index), otherwise 0. P_{siam} and P^{n}_{seg} are the predicted similarity by the DG-Siamese-Net and the segmentation map by the nth baseline-network, respectively. In addition, N indicates the number of classes of density (see Section III-B), and θ_n is the trainable parameter of the nth baseline-network. Equations (6) and (7) are utilized for training a guider and baseline-networks, respectively. Finally, the DG-Net is optimized using the following equation:

$$\mathcal{L}(I^{(i)}, G^{(i)}, \theta) = \mathcal{L}_{\text{siam}}(I^{(i)}) + \sum_{n=1}^{N} \mathcal{L}_{\text{seg}}(I^{(i)}, G^{(i)}, \theta_n). \tag{8}$$

E. Prediction by DG-Net

In the prediction phase, as illustrated in Fig. 1, the following sequential steps are carried out: DG-Net recognizes the density (*S.I.*) of the target objects in an input image ($I^{(i)}$) with the following equation:

$$S.I.(I^{(i)}) = \underset{n}{\text{argmax}} \; s\left(I^{(i)}, R_n\right). \tag{9}$$

The guider then selects the baseline-network corresponding to a density index [$N' = S.I.(I^{(i)})$], and DG-Net generates a semi-predicted segmentation map ($P'^{(i)}$) as follows:

$$P'^{(i)} = P^{N'}_{\text{seg}}(I^{(i)}). \tag{10}$$

With the semi-predicted segmentation map, the density of target objects in the input images can be calculated, and then DG-Siamese-Net is fine-tuned as follows:

$$\mathcal{L}'_{\text{siam}} = \sum_{i} d_{\text{idx}}(P'^{(i)}) \cdot \log P_{\text{siam}}(I^{(i)}, R_n) \tag{11}$$

where $d_{\text{idx}}(P')$ is the one-hot labeled vector of the density, which is calculated from $P'^{(i)}$. Note that, the baseline-network of DG-Siamese-Net has the same architecture as well as shares parameters with the selected baseline-network, and thus the baseline-network of the baseline-network is also fine-tuned, simultaneously, in the fine-tuning step of the DG-Siamese-Net. DG-Net then generates the final prediction ($P^{(i)}$) by using the fine-tuned baseline-network.

IV. Experimental Results

A. Dataset Description

In the evaluations, various datasets, such as aerial imagery datasets [34], [35], [36] were employed. Since the density of the objects should be a crucial feature for aerial imagery, RS datasets are employed to illustrate the strength of DG-Net

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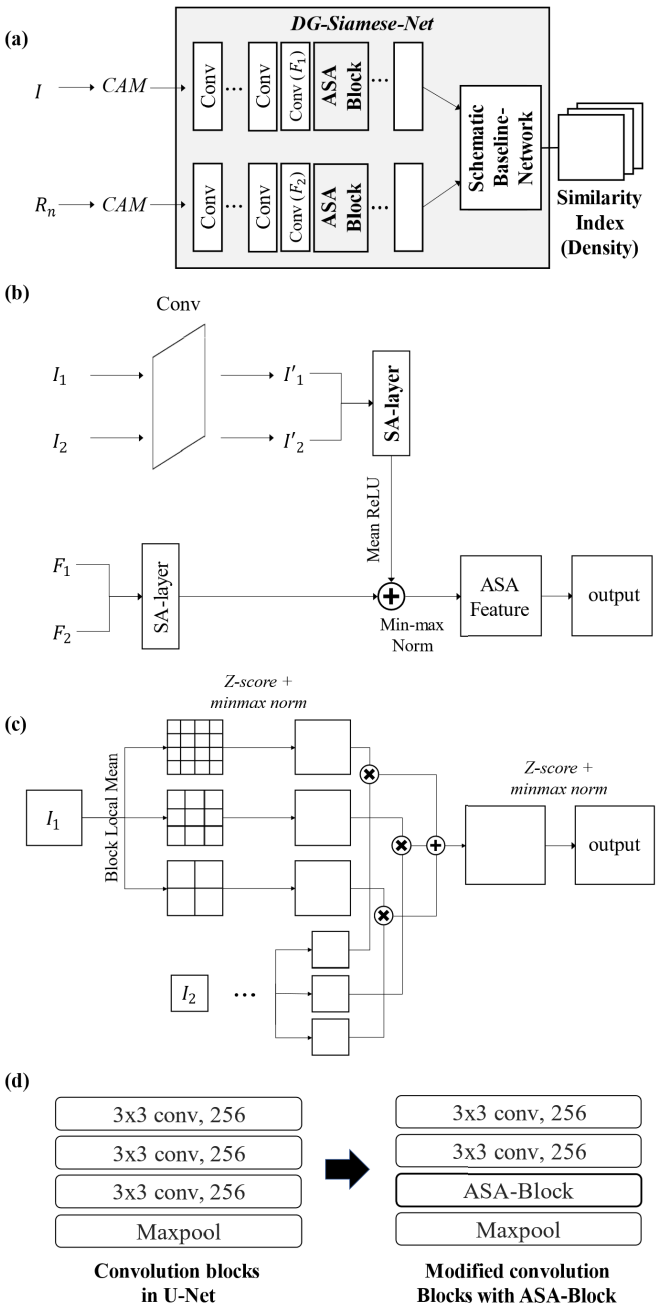

Fig. 2. Detailed architecture of the guider. (a) Schematic pipeline of the guider including DG-Siamese-Net with an ASA-block. (b) Schematic pipeline of the ASA-blocks with SA-layers. The ASA-block generates a similarity-based attention map (ASA-Features). (c) Schematic pipeline of the SA-layer. The SA-layer measures the similarity between two input images. (d) Modified convolution blocks, in which a 3 × 3 convolution block is replaced by an ASA-block, in baseline-network.

effectively. The Inria dataset covers an area of 405 km^2. The aerial RGB-colored image has a spatial resolution of 0.3 m and covers five categories (Austin, Chicago, Kitsap, Tyrol, and Vienna). We randomly cropped the training image into the size of 256 × 256 from the size of 5000 × 5000, and 144 000 images were utilized for each category. The density of buildings was equally distributed for each city.

From the WHU Building dataset, we used one of Satellite dataset I. The dataset was collected from cities all over the world and from various RS resources including QuickBird, Worldview series, IKONOS, and ZY-3 [35]. The WHU dataset contains 204 images of which size is 512 × 512 but randomly cropped so that a total number of 20 400 images are constructed for the WHU dataset. Here, the resolutions of the WHU dataset vary from 0.3 to 2.5 m.

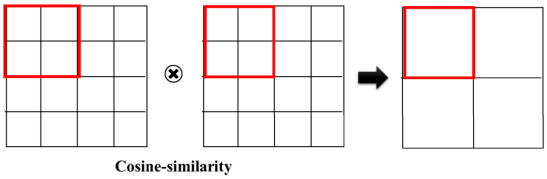

Fig. 3. Detailed description of block local mean. The block local mean operation gets two inputs and generates the similarity-distribution map between the two inputs.

The Massachusetts Building dataset consists of 151 aerial images of the Boston area. The size of the images is 1500 × 1500 pixels for an area of 2.25 km^2, and thus the entire dataset covers roughly 340 km^2. The data was split into a train set of 137 images, a test set of ten images, and a validation set of four images [36]. Here, we randomly cropped the original images to the small images of which size was 256 × 256, and a total of 16 577 images were utilized to the Massachusetts dataset. In the Massachusetts Building dataset, the target maps were obtained by rasterizing building footprints obtained from the OpenStreetMap project. The Massachusetts Building dataset is a large amount of high-quality building footprint data. The dataset covers mostly urban and suburban areas, including buildings of various sizes, individual houses, and garages, which are included in the labels [36].

B. Experimental Setup and Implementations

To demonstrate the segmentation performance of DG-Net, three groups of DL models are utilized as shown in the followings; 1) baseline models including the early vanilla models of FModel, SegNet [18], DeconvNet [15], U-Net [16], and FusionNet [17]; 2) multipath models for the segmentation task including SK-U-Net [26], LADDERNET [24], MPDNet [25], and RefineNet [23]; and 3) SotA models for each category of the benchmark. For instance, S2EPC [10], BSNet [8], TTAS [11], and DIGA [12] are used for the RS benchmark datasets. Note that the SoTA models, used here, exhibit the best performance for the benchmark datasets. In addition, various versions of DG-Nets are constructed with Baseline and SotA models as the baseline-network of DG-Nets. However, the multipath models are utilized as the comparative models, but they are not utilized as the baseline-network of DG-Net because of the memory inefficiency caused by the nested networks.

Here, to implement models, the same parameters were utilized in the experiments. The mini-batch was 12 with the Adam optimizer and batch-normalization [37]. The SotA models and DG-Net were implemented using the public platform TensorFlow [38]. The server for the experiments had two CPUs (E5-2640 v4), four GPUs (NVIDIA TitanXp), and 128GB of RAM. In addition, we applied a k-fold

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TABLE I
SUMMARY OF KEY FEATURES OF SEGMENTATION MODELS

Models	Vanilla Models	Multi-Path Models	SotA Models
FCN	✓		
SegNet	✓		
DeconvNet	✓		
U-Net	✓		
FusionNet	✓		
SK-U-Net		✓	
LADDERNet		✓	
MDPNet		✓	
RefineNet		✓	
S2EPC			✓
BSNet			✓
TTAS			✓
DIGA			✓

cross-validation method to evaluate the proposed model ($k = 5$) [39]. To this end, all datasets were divided into five splits, three splits were used as a train set, another split was used as a validation-set, and the other split was used as a test set. Furthermore, the reference images (R_i) were selected from the train set to prohibit the overfitting problem. For instance, if the reference images are from the test set in the training phase, the model learns the features of the test set and is biased, and thus it could exhibit high performance on the test set. Therefore, the reference images were selected from the train set rather than the validation-sets and test sets.

Furthermore, it is obvious that the duplicated stacked network without fine-tuning could provide strong performance. In contrast, in this study, we demonstrate DG-Net, a novel fine-tuned network comprising a compact stack of layers, to achieve both memory efficiency as well as high performance. To compare the performance of the stacked network and that of DG-Net, five different DG-Nets were implemented and compared, including DG-Net-1, DG-Net-3, DG-Net-10, DG-Net-3 Stacked, and DG-Net-10 Stacked. In the DG-Net-N, N numbers of baseline networks (fine-tunable) were implemented in DG-Net. In contrast, in DG-Net-N Stacked, N numbers of baseline networks (not fine-tunable) were implemented in DG-Net. To demonstrate the feasibility of DG-Nets, all hyperparameters in the constructed models are used as the default values without hyperparameter-tuning to search for the best performance.

To demonstrate the effectiveness of DG-Net in the segmentation tasks, three groups of DL models were employed, such as the vanilla models, multipath models, and SotA models for the RS datasets. The vanilla models were employed as a control group, and multipath models were employed since DG-Net could be considered as a multipath model. In addition, the SotA models for the RS datasets were employed to demonstrate the outstanding performance of DG-Net.

The detailed description are shown in Table I; **1) vanilla models** including the vanilla models of FCN [14], SegNet [18], DeconvNet [15], U-Net [16], and FusionNet [17]; **2) multipath models** for the segmentation task including SK-U-Net [26], LADDERNET [24], MPDNet [25], and RefineNet [23]; **3) SotA models** for each category of the benchmark. For instance, S2EPC [10], BSNet [8], TTAS [11], and DIGA [12] were employed for the RS datasets. In addition, various versions of DG-Nets were implemented using vanilla and SotA models as the baseline. Here, the multipath model was not used as the baseline due to the memory inefficiency caused by the nested architectures. However, the multipath models were employed as the comparative models.

For the quantitative analysis, precision, recall, and IoU values were obtained by using the DG-Net and the associated SotA models. We first measured the evaluation metrics on vanilla (five models), multipath (four models), and SotA models (four models), a total of 13 models. Then, a total of 18 DG-Net-Stacked models (DG-Net-3 and −10 Stacked) with vanilla (five models) and SotA (four models) models were evaluated. Finally, a total of 27 DG-Net models (DG-Net-1, −3, and −10) with vanilla (five models) and SotA (four models) models were evaluated. As a result, 40 models with fivefold-cross validations were evaluated in the experiments using the RS datasets, respectively. To effectively illustrate the performance of various models, the representative performance from each group (DG-Net, DG-Net-Stacked, vanilla, multipath, and SotA) was exhibited by averaging the performances in each group, respectively.

C. Comparative Study

Table II illustrates comprehensive benchmark results of DL models evaluated across various datasets with our DG-Net models. The evaluation metrics employed include IoU, precision, and recall, elaborated as mean values accompanied by their corresponding standard deviation values in parentheses. The experimental results explicitly lead to comprehending the tendencies of DL models and facilitate an understanding of evaluation distributions. The experimental results exhibit the outstanding performance of our DG-Net-3 and DG-Net-10, demonstrating superior advancements over established models in the segmentation. In particular, the DG-Net-3 architecture demonstrates superior improvement, achieving an enhancement of 17.67% in IoU compared to the baseline model, specifically on the Inria dataset. Similarly, the DG-Net-10 architecture exhibits a noteworthy performance elevation, showcasing an improvement of 16.08% in IoU relative to the baseline on the Inria dataset. Compared to the SotA models, the DG-Net-3 model surpasses conventional benchmarks, highlighting an improvement of approximately 5.96% in IoU when contrasted with the existing SotA model for the Inria dataset. Similarly, the DG-Net-10 model demonstrates a significant rise of 4.54% in IoU compared to the SotA on the Inria dataset.

In addition, empirical analysis reveals the remarkable optimality of the DG-Net-3 architecture. Despite its conceivable architectural simplicity or reduced numbers of parameters, the DG-Net-3 model exhibits efficacy, closely approximating the performance of the DG-Net-10 model, which is ostensibly more intricate with many parameters. Such empirical evidence underscores the viability of DG-Net-3, elucidating that considerable performance enhancements can be attained with this variant, all while maintaining marginal performance differences with its more complex counterpart,

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TABLE II

REPRESENTATIVE ANALYSIS OF DG-NET COMPARED TO OTHER DL MODELS. COMPARATIVE EVALUATION OF DL MODELS ACROSS MULTIPLE DATASETS USING THE METRICS OF IOU, PRECISION, AND RECALL. FOR EACH GROUP, THE BEST-PERFORMING MODEL IS DEPICTED. RESULTS ARE PRESENTED AS MEAN VALUES WITH STANDARD DEVIATIONS IN PARENTHESES. THE BEST RESULTS FOR EACH DATASET AND METRIC ARE HIGHLIGHTED AS **BOLD**

	Baseline	Multi-Path	SotA	DG-Net-3	DG-Net-10
Inria	FusionNet	LADDERNet	BSNet	DG-Net-3 (S2EPC)	DG-Net-10 (S2EPC)
IoU	64.93% (1.19)	64.98% (1.04)	72.10% (0.84)	**76.40% (0.98)**	75.38% (1.05)
Precision	89.40% (2.19)	89.12% (1.41)	90.75% (1.43)	90.53% (1.22)	**91.08% (1.30)**
Recall	70.38% (1.27)	70.60% (1.51)	77.83% (1.11)	**83.05% (1.21)**	81.41% (1.67)
Mass-Building	SegNet	SK-U-Net	BSNet	DG-Net-3 (S2EPC)	DG-Net-10 (BSNet)
IoU	59.92% (1.32)	61.45% (1.46)	65.67% (0.85)	68.65% (0.81)	**70.42% (0.62)**
Precision	86.30% (2.21)	88.74% (2.64)	89.62% (1.70)	89.73% (1.66)	**90.83% (2.41)**
Recall	66.27% (1.86)	66.71% (2.09)	71.10% (1.12)	74.55% (1.76)	**75.88% (1.82)**
Mass-Road	U-Net	SK-U-Net	BSNet	DG-Net-3 (BSNet)	DG-Net-10 (BSNet)
IoU	66.27% (0.94)	66.39% (1.31)	73.16% (0.85)	75.46% (1.16)	**78.09% (0.92)**
Precision	89.26% (2.49)	89.50% (1.68)	90.91% (1.40)	91.29% (1.74)	**91.91% (1.73)**
Recall	72.07% (1.54)	72.01% (1.26)	78.95% (1.11)	81.37% (2.18)	**83.89% (1.72)**
WHU	U-Net	MPDNet	BSNet	DG-Net-3 (BSNet)	DG-Net-10 (S2EPC)
IoU	61.15% (0.39)	61.11% (1.50)	65.49% (0.85)	**71.59% (1.16)**	70.67% (1.05)
Precision	88.14% (2.35)	89.24% (2.16)	89.58% (1.66)	**90.69% (1.87)**	90.33% (1.46)
Recall	66.67% (1.09)	66.03% (2.41)	70.91% (1.13)	**77.32% (2.17)**	76.49% (1.68)

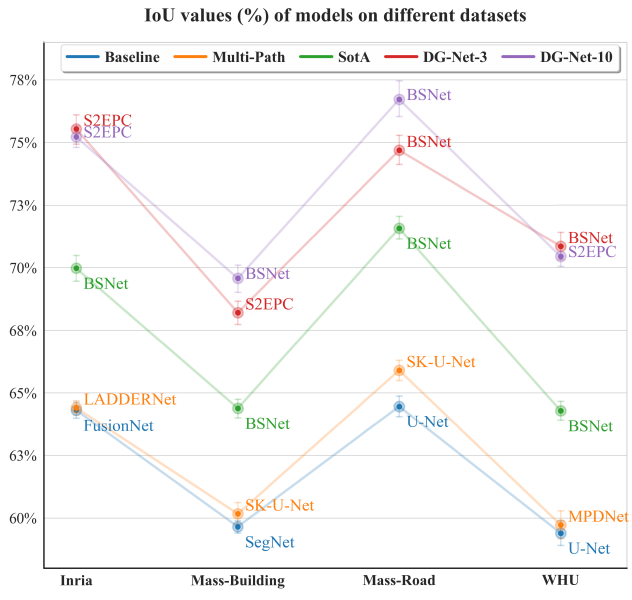

Fig. 4. IoU values obtained by using each model on RS datasets.

DG-Net-10. In summary, Table II and Fig. 4 depict the capabilities and outstanding performances of the DG-Net models, particularly the DG-Net-3 and DG-Net-10, as achieving the SotA performance, efficiency, and optimality compared to other DL models. Moreover, the novel fine-tuning methodology of DG-Net brings out significantly precise segmentation maps, particularly evident in its ability to delineate object boundaries accurately.

Fig. 5 illustrates the representative segmentation results obtained using the best models in the groups of vanilla, multipath, SotA, DG-Net, and DG-Net-Stacked (DG-Net-S). In Fig. 5, the predictions, which were obtained by using DG-Net, showed precisely segmented results. In contrast, the images, which were obtained by using other models, exhibited many incorrect predictions for object segmentation. In the Inria dataset, the vanilla model (FusionNet) showed the mispredicted buildings in the background and imprecise boundary segmentation results. The multipath model (LADDERNet) showed the blurred building prediction due to its ensemble structure. In addition, the SotA model (TTAS) exhibited more precise building boundaries than the vanilla and multipath models but provided the mispredicted buildings in the background. On the contrary, DG-Net with the S2EPC model and DG-Net-10-Stacked with the TTAS model provided the precise predictions than other compatible models in a simple building prediction. Similar trends were exhibited in other datasets of WHU and MASS datasets. In the WHU dataset, vanilla (SegNet), multipath (MPDNet), and SotA (S2EPC) models provided imprecise predictions with empty holes in buildings and mispredictions in the backgrounds than DG-Net with S2EPC and DG-Net-10-Stacked with HMA-Net models. In the MASS building and road dataset, vanilla (FusionNet), multipath (SK-U-Net), and SotA (S2EPC) models provided lump buildings and cut roads, but DG-Net with S2EPC and DG-Net-10-Stacked with BSNet models provided the precise predictions on images of dense buildings and thin and sparse roads. These results demonstrated that DG-Net outperformed the other SotA models in binary segmentation.

Fig. 6 depicts the IoU values attained by diverse DL models and our DG-Net models using the Inria dataset. The systematic arrangement facilitates a cogent juxtaposition, as models are categorized based on incremental IoU values within each cohort. As illustrated in Fig. 6, the DG-Net-3

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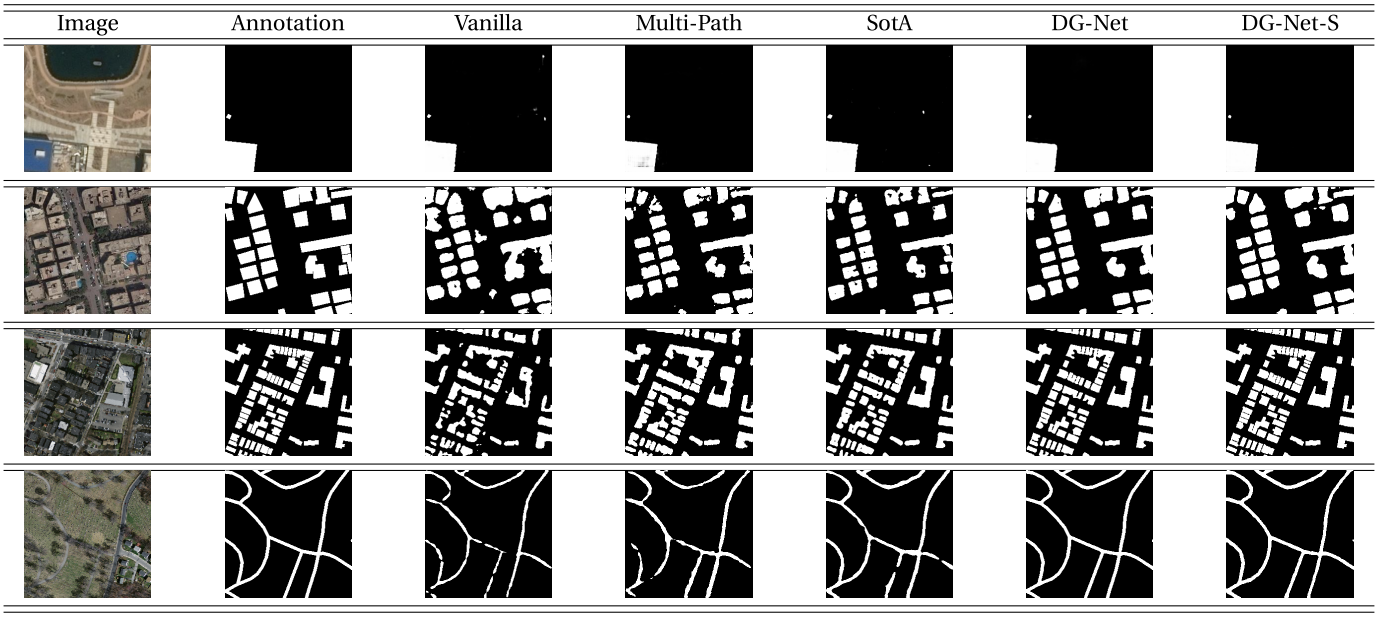

Fig. 5. Representative results of baseline, multipath, SotA, and DG-Net using each dataset. Rows indicate Inria, WHU, Massachusetts Dataset (Building), and Massachusetts Dataset (Road), respectively.

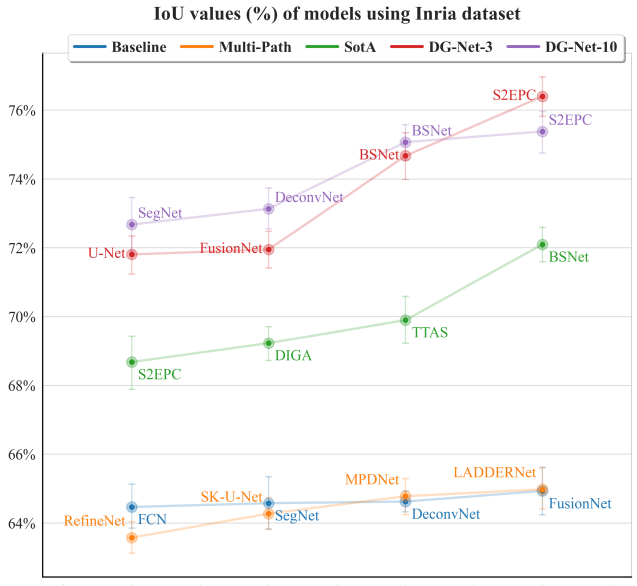

Fig. 6. IoU values obtained by using each model on Inria datasets. DL models are sorted with the increasing IoU values within each group. The best-performing baseline-model for DG-Net is illustrated.

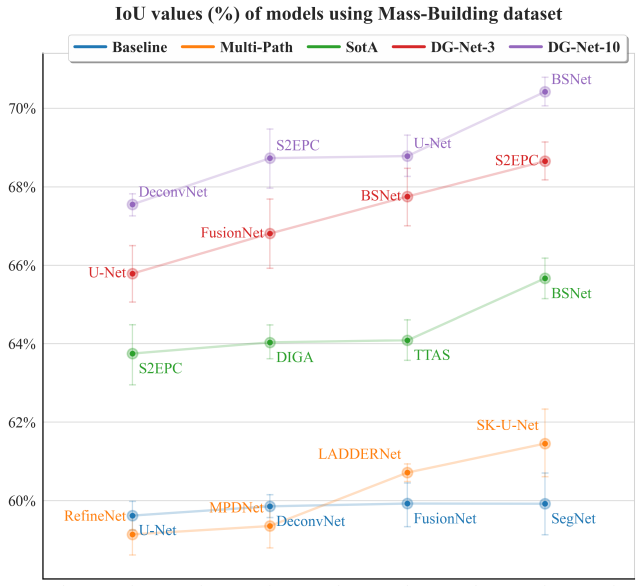

Fig. 7. IoU values obtained by using each model on Inria datasets. DL models are sorted with the increasing IoU values within each group. The best-performing baseline-model for DG-Net is illustrated.

model showcases a significant enhancement of 11.47% in IoU relative to the baseline model. This augmentation highlights the proficiency in effectively addressing the intricacies inherent to the Inria dataset. Similarly, the DG-Net-10 iteration also attains a significant performance improvement, manifesting an increment of 10.45% in IoU compared to the baseline models. Fig. 6 additionally demonstrates the outstanding improvement of DG-Net models against the SotA models. Remarkably, the DG-Net-3 and DG-Net-10 models yield a significant improvement of 4.30% and 3.28% in IoU compared to SotA models. These results emphasize the substantial advancements achieved by our DG-Net models on the Inria dataset.

Fig. 7 presents the IoU values obtained from various DL models and our DG-Net models on the Mass-Building dataset. The systematic arrangement allows for a clear comparison, with models grouped according to incremental IoU values within each category. As depicted in Fig. 7, the DG-Net-3 model demonstrates a significant enhancement of 8.75% in IoU compared to the baseline model. This improvement underscores the model's effectiveness in addressing the complex characteristics inherent to the Mass-Building dataset. Similarly, the DG-Net-10 iteration achieves a substantial performance boost, showing an increase of 8.77% in IoU compared to the baseline models. Notably, Fig. 7 highlights

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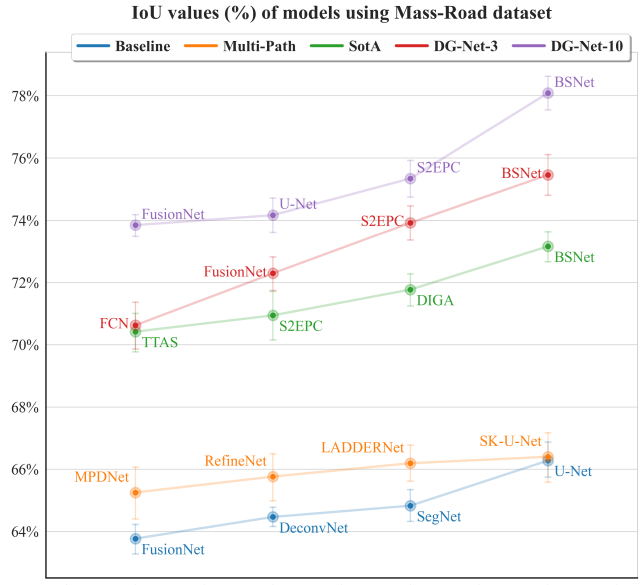

Fig. 8. IoU values obtained by using each model on Inria datasets. DL models are sorted with the increasing IoU values within each group. The best-performing baseline-model for DG-Net is illustrated.

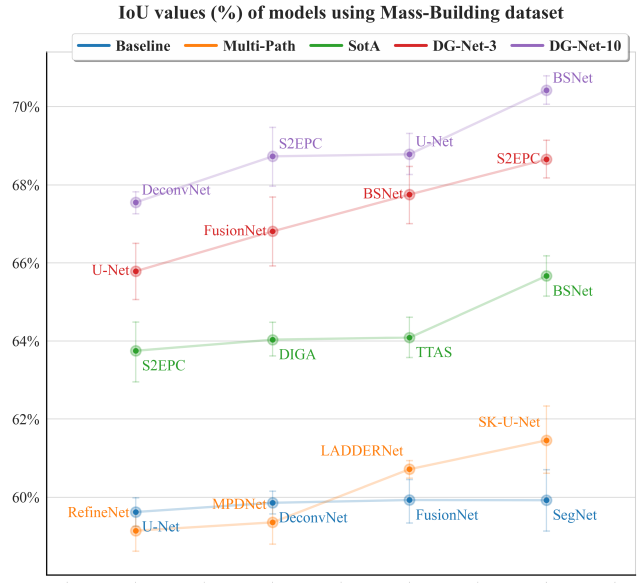

Fig. 9. IoU values obtained by using each model on Inria datasets. DL models are sorted with the increasing IoU values within each group. The best-performing baseline-model for DG-Net is illustrated.

the remarkable advancement of DG-Net models over the SotA models. Particularly, both the DG-Net-3 and DG-Net-10 models achieve impressive enhancements of 3.75% and 4.75% in IoU, respectively, when compared to the SotA models. These results underscore the remarkable performance gains of our DG-Net models on the challenging Mass-Building dataset.

Fig. 8 illustrates the IoU values obtained from various DL models and our DG-Net models on the Mass-Road dataset. The systematic arrangement allows for a clear comparison, grouping models by their incremental IoU values within each category. As evidenced in Fig. 8, the DG-Net-3 model exhibits a substantial increase of 9.19% in IoU compared to the baseline model. This improvement underscores the adeptness in addressing the intricacies inherent in the Mass-Road dataset. Similarly, the DG-Net-10 iteration achieves significant performance growth, showing an augmentation of 11.93% in IoU compared to the baseline models. Notably, Fig. 8 underscores the remarkable progress of DG-Net models over the SotA models. Particularly, both the DG-Net-3 and DG-Net-10 models demonstrate substantial improvements of 2.30% and 4.93% in IoU, respectively, when compared to the SotA models. These results highlight the exceptional performance enhancements of our DG-Net models on the challenging Mass-Road dataset.

Fig. 9 shows the IoU values achieved by various DL models and our DG-Net models on the WHU dataset. The systematic arrangement enables a clear contrast, categorizing models based on their incremental IoU values within each cohort. As indicated in Fig. 9, the DG-Net-3 model demonstrates a remarkable boost of 10.44% in IoU relative to the baseline model. This advancement underscores the proficiency in effectively handling the complexities inherent to the WHU dataset. Similarly, the DG-Net-10 iteration also achieves significant performance enhancement, showcasing an increment of 9.30% in IoU compared to the baseline models. Moreover, Fig. 9 highlights the exceptional progress of DG-Net models over the SotA models. Notably, both the DG-Net-3 and DG-Net-10 models exhibit substantial improvements of 6.10% and 5.17% in IoU, respectively, when compared to the SotA models. These findings underscore the significant performance gains of our DG-Net models on the challenging WHU dataset.

D. Ablation Study

Clearly, N-stacked DG-Net models can provide higher accuracy and IoU values than the normal DG-Net models, which are fine-tuning networks. However, memory efficiency should also be considered while implementing DL models in a real-world environment. Therefore, to measure the accuracy and memory efficiency of DG-Net-N and DG-Net-N Stacked models, the ablation study was conducted. To this end, we implemented the different versions of DG-Nets, of which baseline-networks are based on vanilla and SotA models. Table III and Fig. 10 illustrate the representative results in terms of IoU values and bars indicating the 95% confidence interval (95% C.I.).

In the ablation study presented in Table III, we conduct a detailed comparison between two variants of the DG-Net architecture: the fine-tuned model and the stacked version. The performance evaluation is carried out on four diverse datasets: Inria, Mass-Building, Mass-Road, and WHU. The IoU values are accompanied by 95% confidence intervals (95% C.I.).

Table III depicts intriguing patterns of the DG-Net-3 models. For instance, on the Inria dataset, the stacked version exhibits an IoU of 76.64% with a confidence interval of 75.94%–77.33%, surpassing the fine-tuned model's IoU of 74.67% (73.52%–75.83%). In contrast, in the Mass-Building dataset, the stacked model achieves an IoU of 68.69%

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TABLE III

ABLATION STUDY TO EVALUATE THE PERFORMANCE OF DG-NET (FINE-TUNING MODEL) COMPARED TO THE DG-NET (STACKED MODEL). THE IOU VALUES ARE DEPICTED WITH 95% C.I. THE BEST-PERFORMING STOA MODELS ARE EMPLOYED AS THE BASELINE-NETWORK FOR EACH DG-NET

Name	Inria	Mass-Building	Mass-Road	WHU
DG-Net-3 [BSNet / Stacked]	76.64% (75.94% ~ 77.33%)	68.69% (67.55% ~ 69.84%)	77.28% (76.60% ~ 77.96%)	68.54% (67.85% ~ 69.24%)
DG-Net-3 [BSNet / Fine-Tune]	74.67% (73.52% ~ 75.83%)	67.75% (66.51% ~ 68.99%)	75.46% (74.30% ~ 76.62%)	71.59% (70.43% ~ 72.74%)
DG-Net-3 [S2EPC / Stacked]	73.08% (72.30% ~ 73.85%)	68.67% (68.05% ~ 69.29%)	76.97% (76.20% ~ 77.75%)	68.07% (66.97% ~ 69.16%)
DG-Net-3 [S2EPC / Fine-Tune]	76.40% (75.42% ~ 77.38%)	68.65% (67.84% ~ 69.46%)	73.92% (72.94% ~ 74.90%)	70.12% (69.27% ~ 70.96%)
DG-Net-10 [BSNet / Stacked]	76.20% (75.47% ~ 76.94%)	71.35% (70.31% ~ 72.38%)	78.30% (77.57% ~ 79.02%)	70.61% (69.88% ~ 71.35%)
DG-Net-10 [BSNet / Fine-Tune]	75.07% (74.16% ~ 75.98%)	70.42% (69.79% ~ 71.04%)	78.09% (77.16% ~ 79.01%)	70.22% (69.31% ~ 71.14%)
DG-Net-10 [S2EPC / Stacked]	74.73% (73.87% ~ 75.60%)	71.70% (70.59% ~ 72.80%)	72.83% (71.97% ~ 73.70%)	70.47% (69.61% ~ 71.33%)
DG-Net-10 [S2EPC / Fine-Tune]	75.38% (74.33% ~ 76.43%)	68.73% (67.46% ~ 70.00%)	75.34% (74.30% ~ 76.38%)	70.67% (69.62% ~ 71.72%)

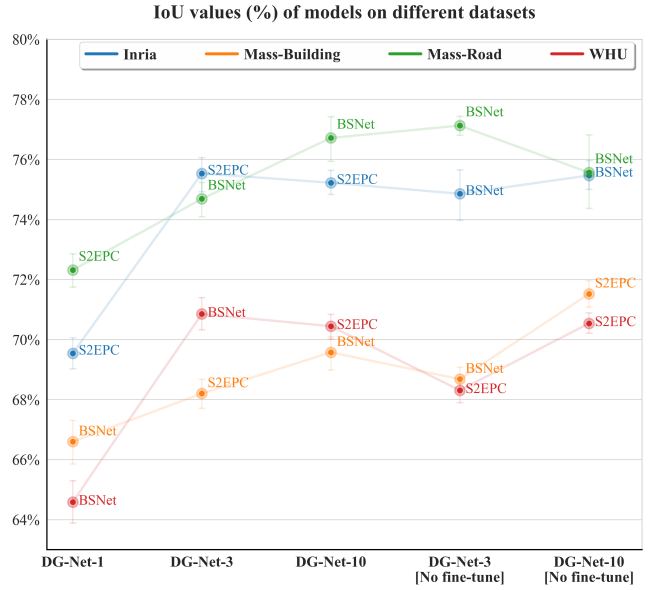

Fig. 10. IoU values obtained by using each model on RS datasets. The best-performing baseline-model for DG-Net is illustrated.

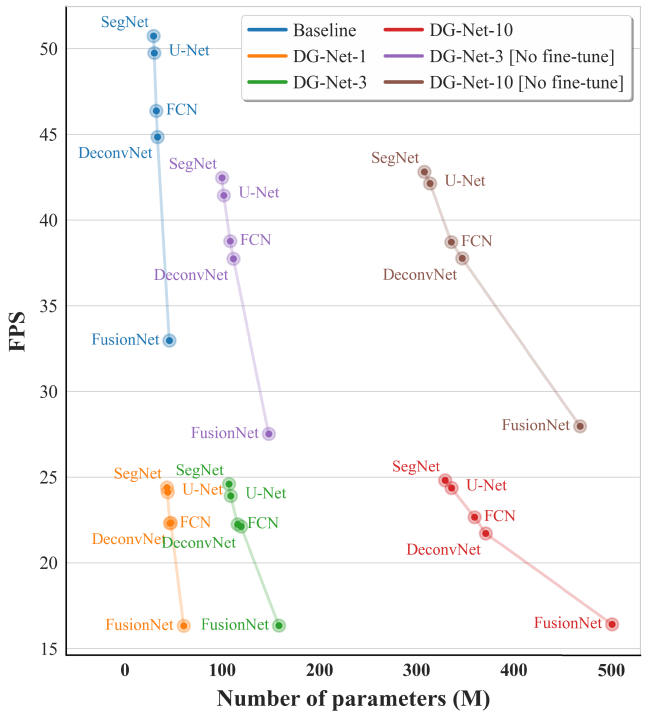

Fig. 11. Comparitive analysis of DG-Nets and other DL-model in the group of *baseline* regarding FPS and number of parameters in DL models.

(67.55%–69.84%), outperforming the fine-tuned model with an IoU of 67.75% (66.51%–68.99%). A similar trend can be observed in the Mass-Road and WHU datasets. Similarly, Table III reveals distinctive performance patterns among the DG-Net-10 models. The stacked variant consistently demonstrates enhanced performance on the Inria dataset with an IoU of 76.20% (75.47%–76.94%), compared to the fine-tuned model's IoU of 75.07% (74.16%–75.98%). Within the context of the Mass-Building dataset, the fine-tuned model slightly surpasses the stacked model with IoU values of 70.42% (69.79%–71.04%) and 70.22% (69.31%–71.14%), respectively. Similarly, in the Mass-Road dataset, the stacked model showcases superior performance with an IoU of 78.30% (77.57%–79.02%), while the fine-tuned model achieves an IoU of 78.09% (77.16%–79.01%). In contrast, both variants exhibit comparable performance within the WHU dataset, each achieving IoU values of 70.61% (69.88%–71.35%) for the stacked model and 70.22% (69.31%–71.14%) for the fine-tuned model.

Fig. 11 shows a comparative analysis of the DG-Net models compared to the *baseline* group, achieving noteworthy observations regarding frames/s (FPS) and the number of parameters. Remarkably, the DG-Net-3 (stack) models demonstrate a commendable similarity in FPS compared to the baseline models since the stacked model costs a simple prediction in the baseline-network with a simple density prediction. The comparison analysis suggests that the stacking mechanism does not substantially compromise the processing speed of the DG-Net-3 models, highlighting its efficiency in handling real-time applications. Similarly, Fig. 12 demonstrates an outstanding comparison analysis of DG-Nets compared to the *SotA* and *multipath* models concerning FPS and the number of parameters. Notably, the DG-Net-3 (fine-tuning) model is underlined as a high-efficiency model

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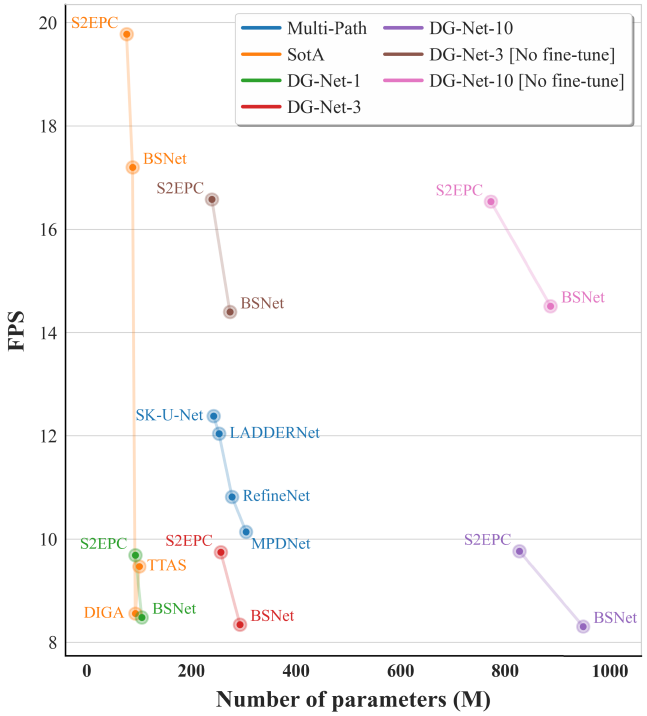

Fig. 12. Comparitive analysis of DG-Nets and other DL-model in the groups of *SotA* and *multipath* regarding FPS and number of parameters in DL models.

in terms of FPS, outperforming test-time adaptation SotA models (TTAS and DIGA). The results imply that the fine-tuning methodology applied to DG-Net-3 leads to a computational advantage with ASA-block, allowing for faster inference times than other SotA techniques. Consequently, Figs. 11 and 12 provide a comprehensive visualization in terms of computational complexity, elucidating the comparative performance of DG-Nets in terms of FPS and parameter counts within different scenarios.

The experimental results in Table III and Figs. 10–12 provide the novel finding that the DG-Net-3 model, which outperforms other SotA models (see Section IV-C), could segment multi objects from RS images with high predictive accuracies compared to the stacked models despite its a few numbers of parameters than the stacked models. Since the average gap between DG-Net and DG-Net stacked models was less than 1.0%, it is acceptable that the DG-Net is comparable with the stacked models, due to its novel fine-tuning pipeline. The experimental results verify that the fine-tuning DG-Net models can be comparable with DG-Net stacked models to provide high performances in the segmentation tasks while achieving memory efficiency. Therefore, the experimental results could be summarized as follows: 1) the DG-Net framework with SotA models could improve the predictive accuracy of SotA models in segmenting objects in aerial imagery. 2) The DG-Net, which is basically a test-time adaptation learning model, could provide SotA segmentation performance in segmenting objects in aerial imagery compared to the ensemble models (stacked DG-Net) with a large number of parameters, due to its novel fine-tuning methodology. 3) In addition, the DG-Net could achieve memory efficiency compared to ensemble models as well as high segmentation performance.

V. Discussion

A. Novelty and Contribution

In this article, we studied a methodology for fine-grained segmentation in RS imagery. As verified in our preliminary study, fine-grained segmentation is hampered by the inherent drawbacks of probability-based segmentation. Particularly, the probability-based segmentation approach prioritizes the inner pixels of geospatial objects in RS imagery, leading to a deterioration in the segmentation performance and an inability to comprehend boundaries in detail. To address this, we mathematically analyzed probability-based segmentation and subsequently proposed a novel methodology to alleviate the limitations, transitioning into density-based segmentation. In addition, we verified the mathematical analysis by conducting the empirical analysis (see Appendix). By initially classifying the input image based on its density, the proposed DL model, DG-Net, could select the appropriate subnetwork (baseline network), thereby leading to the improvement in biased optimization.

Furthermore, to implement density-based segmentation, we designed the DG-Net and the DG-Siamese-Net, which is an embedded subnetwork. Furthermore, we developed the novel ASA-block and SA-Layer to enhance the performance of density prediction significantly. The design of these modules allowed the DG-Net to achieve superior segmentation performance compared to other SotA models in fine-grained segmentation. Consequently, we significantly anticipate that the novel modules will be incorporated into other models, significantly intensifying the performance of DL models in fine-grained segmentation tasks within RS imagery.

B. Limitations and Future Works

However, in addressing the scope of our research, several limitations and future work should be considered. First, we applied the DG-Net with an exclusive focus on binary segmentation. However, the noticeable necessity to extend our methodologies should be required to incorporate multicategorical segmentation, broadening the applicability and generalizability of our techniques. In particular, the model's capacity for generalization holds significant importance. In our prior research, we introduced a self-mutating network designed for domain adaptive segmentation in aerial imagery. Notably, the pivotal module from the self-mutating network has the potential to be integrated into DG-Net. Second, we have successfully demonstrated the remarkable fine-grained segmentation capabilities of our module when integrated into other cutting-edge models. However, the optimal combination of our module with cutting-edge models is further investigated for better segmentation. Moreover, the limitations stemming from the modality constraints within our study are evident, particularly when considering the integration of hyperspectral imagery in RS applications. As such, it is imperative to chart the course of future research toward accommodating various modalities for broader segmentation

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applications. Finally, numerous challenges persist due to the intrinsic attributes of RS imagery, such as the presence of noise, degradation, and inherent variability. These challenges may cause domain shifts or gaps, underscoring the need for continued investigation in upcoming endeavors.

VI. Conclusion

In conclusion, we proposed a novel framework, called a DG-Net, to improve the performance of fine-grained segmentation by jointly using SotA models. In addition, we mathematically and empirically studied the difference between probability- and density-based segmentation. DG-Net selects the optimal baseline-network among candidate baseline-networks relying on the density of the target objects in input images and fine-tunes the selected baseline-network. By the devised fine-tuning method, the DG-Net achieved high performance and memory efficiency. The experimental results demonstrated that the DG-Net significantly enhanced the fine-grained segmentation performance. In addition, DG-Net required a small number of parameters to calculate it compared to the multipath and stacked networks. The main contributions of this study are the development of fine-tuning methods, including an ASA-block and an SA-layer to improve an IoU, precision, and recall values, which are the key metrics for a segmentation task, and the adaptive fine-tuning method to achieve memory efficiency. Note that, the currently proposed novel framework can be applied to any other advanced SotA models for more enhanced segmentation without complex implementations. However, finding the best SotA models and hyperparameters is required for further enhancement in a segmentation task. The associated study has remained a future work.

Appendix

Preliminary Study

Definition 1 (Probability-Based Segmentation): Let M be a function that contains a DL model designed for performing segmentation tasks. Let $I \in \mathbb{R}^{H,W,3}$ denote an RGB image and $G \in \mathbb{R}^{H,W,C}$ be the corresponding ground truth, where H and W indicate the height and width, respectively, and C represents the number of categories. Then, the prediction generated by M is denoted as $M(I) \in \mathbb{R}^{H,W,C}$. When the last layer of M employs the softmax function, and the training of M employs an entropy-based loss function, the resulting system is formally defined as probability-based segmentation.

For instance, suppose the cross-entropy function, which is $\mathcal{L}_{\text{ce}}(y, \hat{y}) = y \log \hat{y}$, and M is trained as below

$$\min \sum_{h,w} \mathcal{L}_{\text{ce}}(G_{(h,w)}, M(I)_{(h,w)}) \tag{12}$$

where h and w denote specific pixel locations within the image. Accordingly, the outcome of the model M can be characterized as a probability-based segmentation by Definition 1. Entropy-based optimization has been prevalently employed in training DL systems due to its simplicity. Nonetheless, the probability-based optimization strategy, applied individually to each pixel, fails to recognize the inherent morphological characteristics of the target objects. Consequently, the probability-based approach tends to yield imprecise predictions in the context of the segmentation task.

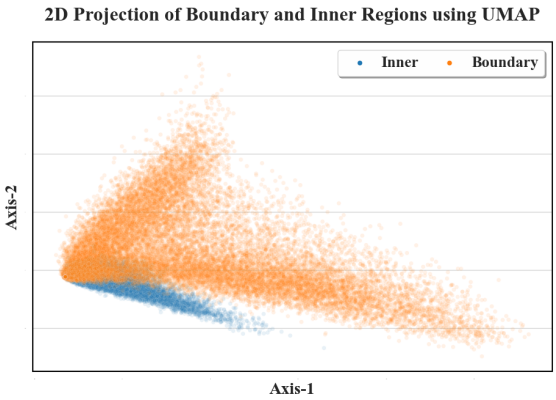

Fig. 13. Similarity of attributes between inner and boundary pixels of targets in RS Imagery on a 2-D plane using UMAP method. Inner pixels of targets in RS imagery exhibit similar attributes compared to boundary pixels. The averaged results were observed in experiments conducted with Inria, WHU, Mass-Building, and Mass-Road datasets.

Conjecture 1: Probability-based segmentation optimizes DL models in a manner where pixels exhibiting similar attributes are prioritized during the optimization process.

By Conjecture 1, the strategy of probability-based segmentation aims to enhance the ability of DL models to identify and capture local patterns and correlations among adjacent pixels, ultimately contributing to more accurate and coherent segmentation outcomes. This strategy of probability-based segmentation can be intuitively comprehended as considering pixels with similar characteristics as easy samples, leading to expediting the entire training process.

Definition 2 (Inner and Boundary Pixels): Let $I \in \mathbb{R}^{H,W,3}$ denote an RS image and $G \in \mathbb{R}^{H,W}$ be the corresponding ground truth as a binary segmentation mask. To detect the boundaries of targets, the boundary detection function f_{bd} (e.g., Sobel and Canny filter) may be employed. Subsequently, the sets of interior ($\mathbb{T}$), boundary ($\mathbb{B}$), and inner ($\mathbb{I}$) are defined as below

$$\begin{aligned} \mathbb{T}(I) &= \{(h,w) | G(h,w) = 1\} \\ \mathbb{B}(I) &= \left\{(h,w) | \Big(f_{\text{bd}}\big(G\big)(h,w) = 1\Big) \wedge \Big(G(h,w) = 1\Big)\right\} \\ \mathbb{I}(I) &= \mathbb{T} - \mathbb{B}. \end{aligned} \tag{13}$$

In this study, we leveraged the canny edge filter and dilation method with several iterations ($n = 3$) to manipulate enough number of boundary pixels for statistics, and thus $f_{\text{bd}} = f_{\text{canny}} \circ (f_{\text{dilation}})^n$.

Proposition 1: Inner pixels of targets in RS imagery exhibit more similar attributes compared to boundary pixels.

Fig. 13 illustrates the distribution of inner and boundary pixels projected onto a 2-D plane using the uniform manifold approximation and projection (UMAP) method [40]. The UMAP projection was applied to the pixels within a window of size 3×3. As depicted in Fig. 13, the inner pixels exhibit a densely clustered pattern, indicating similar characteristics,

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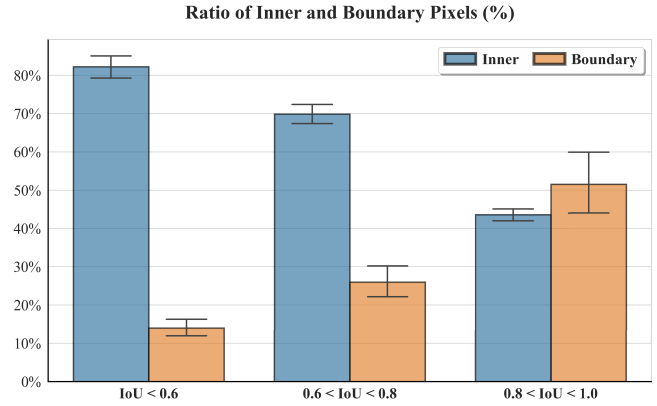

Fig. 14. Ratio of inner and boundary pixels in the area where the gradient significantly decreases during DL model training via probability-based segmentation. The averaged results were observed in experiments conducted with the U-Net and FusionNet models on Inria and WHU datasets. Error bar refers to 95% confidence interval.

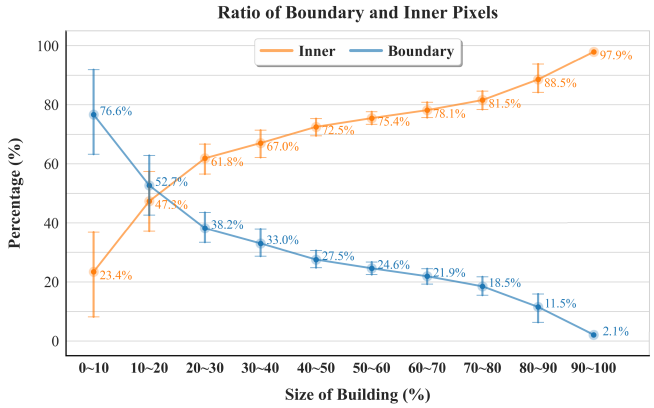

Fig. 15. Variation of inner and boundary pixel ratios with building size. The building size ratio is computed relative to the size of a single image. Averaged results were obtained from experiments using Inria, WHU, and Mass-Building datasets. Error bars represent the 95% confidence interval.

while the boundary pixels are widely scattered, implying diverse characteristics.

Lemma 1: In the context of RS imagery, probability-based segmentation prioritizes the optimization of inner pixels within target objects rather than their corresponding boundaries. Consequently, within the framework of probability-based segmentation, the inner pixels are categorized as easy samples, while the boundary pixels are categorized as hard samples.

Based on Conjecture 1 and Proposition 1, it can be supposed that the optimization process with DL models prioritizes the inner pixels of target objects in RS imagery over the boundary pixels, resulting from the attribute that inner pixels exhibit more consistent characteristics than boundary pixels. Consequently, it is reasonable to conjecture that during the initial stages of training with low accuracy, the training strategy focuses on optimizing the inner pixels, while in the later stages with high accuracy, the emphasis shifts to optimizing the boundary pixels.

In previous studies [31], [41], [42], the concepts of hard and easy samples within a field of DL have been delineated. A hard sample is characterized as a data instance with which DL models grapple, encountering difficulties in accurate classification or prediction. Such challenges are often attributable to the boundary pixels, which are considered hard samples owing to their pronounced variations and irregularities. Conversely, the inner pixels, exhibiting more consistent and uniform characteristics, are categorized as easy samples.

As shown in Fig. 14, probability-based segmentation prioritizes the optimization of inner pixels in target objects, especially when the IoU values are below 60%, as evidenced by the significant decrease in gradients ($>$ 80%) for inner pixels during DL model training. However, as the IoU values reach high levels ($>$ 80%), the focus of the optimization strategy shifts, with boundary pixels being emphasized more than inner pixels, indicating a sequential learning process where inner pixels are learned first rather than boundaries.

Proposition 2: In RS imagery, the number of inner and boundary pixels of the target object increases with the number of target objects in a single image. However, the increment is much greater for the inner pixels than for the boundary pixels.

Particularly, Proposition 2 is intuitively carried out from an increasing number of individual target objects in RS imagery. As the number of target objects increases within a single image, the cumulative count of inner pixels also increases, reflecting the expansion of the target object. Consequently, while both inner and boundary pixels exhibit an increase, the growth is substantially more powerful for the inner pixels than for the boundary pixels.

In Fig. 15, it can be observed that the variation in the ratio of boundary and inner pixels with increasing building size. For small buildings, the portion of boundary pixels is notably larger compared to inner pixels (76.6% versus 23.4%). However, as the building size increases, the ratio of inner pixels exponentially increases (97.9% versus 2.1%), providing empirical evidence to support Proposition 2.

Definition 3 (Density): Let $I \in \mathbb{R}^{H,W,3}$ denote an RS image and $G \in \mathbb{R}^{H,W}$ be the corresponding ground truth as a binary segmentation mask. Let an area ($\mathbb{A}$) of a target in a single image (I) be defined as below

$$\mathbb{A}(I) = \frac{n\big(\mathbb{T}(I)\big)}{HW} \tag{14}$$

where $n(A)$ is the number of elements in set A. Subsequently, a unit area ($\mathbb{A}_u(I)$), which is an area ($\mathbb{A}$) per target object, is defined as $\mathbb{A}_u(I) = \mathbb{A}(I)/\mathbb{N}$, where $\mathbb{N}$ is a number of target object. In this study, we defined the unit area as *density* (d_I), and thus the density of I is defined as below

$$d_I = \frac{1}{HW} \cdot \frac{n\big(\mathbb{T}(I)\big)}{\text{number of target object}}. \tag{15}$$

Definition 4 (False Pixel Distance (FPD) and Mean FPD (mFPD)): False positive distance (FPD, $\mathbb{D}$) is defined as a ratio of the Euclidean distance between a mispredicted pixel and the center of an individual target object to the area of the target object. Suppose $\mathbb{T}(G_i)$ represents an ith individual instance in image I and ground truth G, and thus

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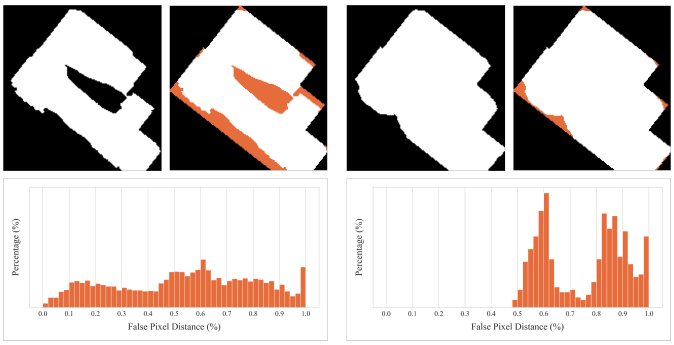

Fig. 16. Samples of FPD visualization. A lower FPD value indicates that the false negatives are primarily concentrated within the inner pixels of the targets. Conversely, a higher FPD value suggests that the errors are located predominantly along the boundary of the targets.

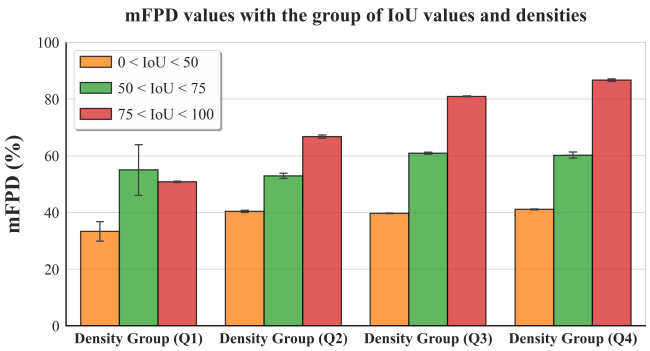

Fig. 17. Relationship between mFPD values and density groups alongside mIoU values during the training of the DL model. The increasing density values correspond to a significant rise in mFPD values. Averaged results were obtained from experiments using U-Net with Inria, WHU, and Mass-Building datasets. Error bars represent the 95% confidence interval.

$\bigcup_i \mathbb{T}(G_i) = \mathbb{G}$. Subsequently, the FPD, denoted as $\mathbb{D}\big(P_i; (h, w)\big)$, at pixel (h, w) for the ith instance in the prediction P, is formulated as follows:

$$\mathbb{D}\big(P_i; (h, w)\big) = \frac{\|(h, w) - \hat{p}_i\|}{n\big(\mathbb{T}(G_i)\big)}$$
$$\text{where}\,\hat{p}_i = \frac{1}{n\big(\mathbb{T}(G_i)\big)} \sum_{p}^{\mathbb{T}(G_i)} p \tag{16}$$

where $\hat{p}_i$ is the center position of the target object, G is the corresponding ground truth, and $\|x\|$ is l_2-norm.

Fig. 16 depicts the distribution and histogram of FPD. In Fig. 16 (left), the mispredictions mainly occur within the inner regions of the building, leading to apparent voids or large holes. Consequently, the FPD histogram exhibits a uniform and even distribution in the inner and boundary of the building. In contrast, Fig. 16 (right) shows that solely the boundary pixels are inaccurately predicted, and thus, the FPD tends to be distributed as larger values, indicating an engagement of erroneous predictions primarily at the boundary of the target.

In addition, we defined mean FPD (mFPD, $\bar{\mathbb{D}}$) to leverage the average false negative rate, and thus the mFPD ($\bar{\mathbb{D}}(P)$) for all instances in P is formulated as below

$$\bar{\mathbb{D}}(P) = \frac{1}{n\big(\mathbb{T}(G)\big)} \sum_{i} \sum_{s}^{P_i'} \mathbb{D}\big(P_i; s\big) \tag{17}$$

where P_i' represents a set of mispredicted pixels in P. Therefore, a high mFPD value (≈ 1) implies a scarcity of precise predictions in boundaries, while a low mFPD value (≈ 0) indicates that the interior region near the center of the building contains voids or empty spaces. In addition, the middle mFPD value ($= 0.5$) indicates a uniform distribution within inner and boundaries.

Lemma 2: In probability-based segmentation, an increase in density brings out an elevation of the mFPD at the same level of IoU value.

Lemma 2: May be carried out, contingent upon prior optimization in the inner pixels over the boundary pixels within the DL models, as defined by Lemma 1, and concomitant with the increase in the number of inner pixels comparative to the boundary pixels, correlated with the increase in the density value of RS imagery (Proposition 2). The preferential optimization of inner pixels, as elucidated in Lemma 1, establishes the correlation between density and inner pixels as a critical aspect. In accordance with Proposition 2, an increase in density is commensurate with a corresponding increase in the ratio of inner pixels to boundary pixels. Such an augmentation in density indirectly results in a more pronounced proportion of inner pixels, thereby contributing to an elevation of the FPD at an equivalent level of IoU value.

In Fig. 17, the interrelation between mFPD values and density groups, along with mIoU values, is explicitly depicted during the training phase of the DL model. Fig. 17 strongly corroborates Lemma 2, illustrating how escalating density values correspond to a significant augmentation in mFPD values at the same level of IoU value. Lemma 2 reinforces the underlying principle that an increase in density, particularly as defined in the context of RS imagery, is concomitant with a proportional increase in mFPD, affirming the critical role of inner pixel prioritization.

Proposition 2: In probability-based segmentation, the difficulty level (easy or hard samples) is influenced by the density of RS imagery.

As the density value increases in RS imagery, there emerges a proportional escalation in the number of inner pixels, which are classified as easy samples for DL models, compared to the boundary pixels, which are categorized as hard samples. Consequently, RS images with vast density are leaned to be classified as easy samples due to the prevailing preponderance of inner pixels over boundary pixels. In contrast, the RS images with either decreased density or containing many buildings are predisposed to be categorized as hard samples. Such a subtle insight into the correlations between density and difficulty levels fosters an enhanced analytical framework within the optimization of the DL models, leading to Proposition 3 as an instrumental concept in refining the probability-based segmentation methodologies.

Lemma 3: In probability-based segmentation, as the learning rate of the DL model increases, the mFPD values also increase.

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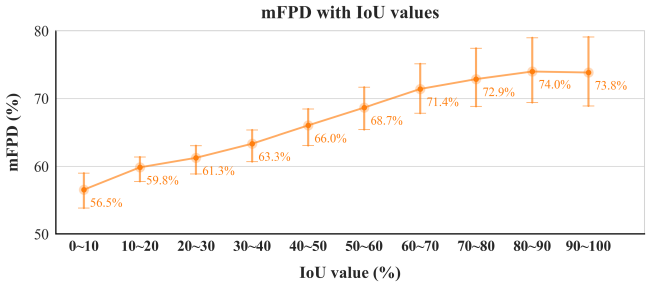

Fig. 18. Relationship between mFPD values and progress of DL models with increasing mIoU values. The increasing IoU values correspond to a significant rise in mFPD values. Averaged results were obtained from experiments using U-Net and U-NetPPL with Inria, WHU, and Mass-Building datasets. Error bars represent the 95% confidence interval.

Lemma 3: Is established via the integration of understandings derived from Proposition 3 and Lemma 1. Within the context of RS imagery, Lemma 1 posits that probability-based segmentation prioritizes the optimization of inner pixels, categorizing them as easy samples, whereas boundary pixels are supposed to be hard samples. Meanwhile, Proposition 3 emphasizes the influence of density in RS imagery on the differentiation between these easy and hard samples. Subsequently, within the optimization process, the model leans to prioritize the training of inner pixels, which are considered easy samples. Consequently, the prioritization within the inner pixels carries out a relative undertraining of the boundary pixels, leading to an elevation of mFPD values. Therefore, Lemma 3 carries out the understanding that the mFPD is continuously increasing within the context of further optimization. In addition, it offers insight into how variations in learning rate impact the segmentation process, thereby further enriching the understanding of the optimization strategies within DL frameworks.

As illustrated in Fig. 17, the mFPD values are increasing alongside the increase in the IoU values in all density groups. Furthermore, Fig. 18 strongly supports Lemma 3 with the increase of mFPD values with the elevation of IoU values while the training of DL models.

Proposition 3: A high value of FPD value signifies a deficiency of fine-grained segmentation.

Proposition 4: Elucidates an important relationship within the field of DL and image segmentation, whereby a high mFPD value represents a deficiency in fine-grained segmentation. The elevation in the FPD value implies an over-generalization in the segmentation process, leading to potential misclassification of boundary pixels of the image. The imprecisions of boundaries indicate the incapacity of DL models to capture intricate details, resulting in a loss of segmentation granularity.

Theorem 1: The sequential learning process in probability-based segmentation hinders the performance in achieving fine-grained segmentation.

By integrating the insights furnished by Lemmas 2 and 3 and Proposition 4, Theorem 1 concentrates on the underlying limitations of probability-based segmentation; lack of fine-grained segmentation. Particularly, Lemma 3 reveals the tendency of increasing mFPD values with the enhancement of the learning rate, indicating an intricate interplay between learning dynamics and segmentation granularity. Meanwhile, Lemma 2 emphasizes the relationship between increased density and the elevation of mFPD and thus prioritizes the optimization of inner pixel prioritization. In accordance with the principles encapsulated in these Lemmas, the sequential learning process debilitates the emphasis on boundary-oriented segmentation, leading to hampering the fine-grained details. The lack of granularity manifests as a limitation, where the prioritization of optimization within inner pixels contributes to a relative neglect of boundary pixels, culminating in the observed deficiency in fine-grained segmentation. Consequently, Theorem 1 emphasizes the need for a more balanced approach within the learning process, facilitating an alignment for fine-grained segmentation.

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